Frontier



[C]Worthy

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

[C]Worthy

Organization location (we welcome applicants from anywhere in the world)

Boulder, Colorado, USA

Name(s) of primary point(s) of contact for this application

Matthew Long

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

Non-Profit Focused Research Organization

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Ocean Alkalinity Process-Model Testbed

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Research context

Advancing ocean alkalinity enhancement (OAE) as a credible CDR pathway requires MRV frameworks that can answer two near-field questions with confidence: (1) What are the alkalinity production rates from mineral feedstocks? (2) What are the thresholds and loss patterns associated with mineral precipitation triggered by alkalinity addition?

When alkalinity is delivered as solid-phase mineral particles (e.g., olivine, calcite), the amount that reaches the ocean depends on dissolution, which in the water column is governed by the balance of sinking and dissolution rates. Material may also accumulate or be deposited on the seafloor, where dissolution yields a bottom-intensified alkalinity flux.

Alkalinity addition—including via electrochemical methods—can also trigger mineral precipitation, removing alkalinity from solution. This loss may be transient: some precipitates redissolve rapidly, and sinking particulates can dissolve at depth, displacing the alkalinity source function downward. But because precipitation is limited by particle nucleation sites, it is also possible to trigger runaway precipitation, in which more alkalinity is lost from solution than was added.

Quantifying these effects is complicated by incomplete process understanding. And while empirical constraints on individual rate processes and thresholds can be established, the relevance of uncertainty in these terms is not obvious without considering the context of the overall physical setting and system uncertainty.

Core research goal

Our primary goal is to advance the capacity to mechanistically simulate the precipitation and dissolution dynamics relevant to OAE. We will pursue this through three integrated objectives:

1. **Mechanistic model.** Develop a process model of particulate sinking, dissolution, and precipitation capable of reproducing patterns observed in laboratory studies and ocean field trials.
2. **Lightweight testbed.** Implement the model in an ultra-lightweight physical configuration that supports rapid prototyping and testing cycles. Reducing physical complexity isolates the biogeochemical processes of interest, enabling cleaner comparisons against observational constraints than are tractable in fully 3D, turbulent configurations.

3. **Uncertainty quantification.** Exploit the low computational cost of the testbed to run large perturbed-parameter ensembles, enabling systematic exploration of parameter space and quantitative assessment of uncertainty in OAE-relevant outcomes.

We will build the testbed as a one-dimensional (1D) OAE process-model configuration within C-Star, [C]Worthy's open-source modeling platform, which couples the Regional Ocean Modeling System (ROMS) with the Marine Biogeochemistry Library (MARBL). ROMS-MARBL will be configured as a 1D column with stratification, vertical mixing, and lateral flushing specified to span representative near-field physical regimes. The testbed will include a generalized representation of size-resolved sinking particulate material, with parameters governing sinking, dissolution, aggregation/fragmentation, and precipitation of mineral feedstocks. Particles created in situ via precipitation provide an additional particle source. Calibration and validation will draw on laboratory and field observations from our partner Ebb Carbon, alongside data from the broader OAE research community.

All process formulations will be implemented directly within ROMS-MARBL, so parameterizations validated in the computationally tractable 1D environment are immediately available in fully explicit 3D regional configurations.

How this advances Frontier's priority innovation area

This project directly addresses *"Improving the MRV toolkit for open-system pathways."* Near-field MRV for OAE requires quantitative assessment of how much alkalinity has effectively been added to the ocean. Advancing modeling capacity in this area will enable robust characterization of how those processes contribute to overall uptake efficiency and its uncertainty—and thereby contribute to effective prioritization of measurement effort where it matters most. For example, uncertainty in the specific threshold triggering precipitation may be unimportant in regions where the precipitate is certain to redissolve. By providing a framework to bound the relevance of outstanding uncertainties, the testbed enables the field to prioritize research investment where it most reduces MRV uncertainty, and to defensibly narrow confidence intervals around crediting claims.

What needs to happen for step-change advancements in cost and scale

Three things must come together:

1. **Empirical constraints must mature in parallel.** Testbed-based parameter calibration must be grounded in observations. The Ebb Carbon partnership anchors this for electrochemical pathways, and we are actively coordinating with the broader OAE community to integrate mineral-feedstock datasets as they become available.
2. **Parameterizations must generalize across deployment contexts.** Mineral dissolution and precipitation kinetics are sensitive to particle size distributions, surface chemistry, and local conditions. Whether a tractable number of parameters can capture this variability across deployment settings is an open question. Systematic uncertainty

analysis will enable us to evaluate where additional process resolution is required versus where simplification is defensible.

3. **Standardization must replace bespoke per-project configurations.** Long-term cost reductions in OAE MRV depend on the field converging on shared, defensible defaults rather than bespoke approaches for each deployment. Modeling is a key tool for this: systematic ensemble analysis with the testbed will identify the parameter regimes where processes are well-characterized and defensible approximations or simplified treatments can suffice, enabling the community to standardize around them with confidence.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

Key research questions

The project is organized around three interlinked research questions that frame near-field MRV challenges for OAE:

1. Under what conditions does alkalinity addition trigger mineral precipitation, what is the resulting alkalinity loss, and how much of that loss is recovered through redissolution at depth?
2. How do dissolution and precipitation processes combine to determine net alkalinity production—and how does that net production vary across representative near-field physical regimes (stratification, mixing, lateral flushing, dosing strategy)?
3. Are there regions in parameter space where processes are well-enough characterized to support defensible simplified treatments?

Approach overview

We answer these questions by building a 1D OAE process-model testbed within C-Star ([C]Worthy's open-source ROMS-MARBL platform), calibrating it against laboratory and field datasets, and using it as the engine for systematic perturbed-parameter ensemble analysis. The testbed is intentionally minimal in physical complexity: a single vertical column with idealized stratification, parameterized lateral flushing, and explicit size-resolved sinking particulates. This reduced-complexity physical configuration isolates biogeochemical processes for clean comparison against observations, and reduces computational cost enough to run the ensembles needed to map uncertainty across the relevant parameter space. However, because all process formulations are implemented natively within

ROMS-MARBL, every parameterization developed and validated in the 1D testbed is immediately portable to fully explicit 3D regional configurations.

The work is organized into six tasks.

Task 1. Configure a generalized 1D physical environment in C-Star. We will configure a 1D ROMS-MARBL column with idealized vertical stratification and parameterized lateral flushing chosen to span the range of representative near-field regimes relevant to OAE deployments—from strongly stratified coastal sites to deep, well-mixed offshore environments. Lateral flushing is parameterized as a relaxation toward a prescribed background state, allowing systematic exploration of how residence time mediates the balance between dissolution, precipitation, and downstream transport. The 1D column model will represent upper-ocean turbulence and mixed-layer dynamics using the standard boundary layer and interior mixing parameterizations already in ROMS.

Task 2. Implement a generalized, size-resolved, explicitly-sinking particulate module. We will develop a MARBL particulate module that represents size-resolved sinking classes with specified sinking velocities and dissolution rates, accommodating both externally-dosed feedstock and particles generated in situ by precipitation. Each size class for each mineral constituent carries parameters for sinking, dissolution kinetics, aggregation/fragmentation (transfer across size classes), and—where applicable—precipitation. The module is designed as a general "chassis": feedstock- and precipitate-specific processes are parameterized distinctly but plug into a shared numerical solution framework. As a component of this work, we anticipate a requirement to implement cations (e.g., Mg^{2+}, Ca^{2+}) in MARBL, which presently tracks just total alkalinity.

Task 3. Develop and validate dissolution and precipitation parameterizations. Process formulations will be developed and validated against empirical datasets spanning laboratory, mesocosm, and field scales. Laboratory constraints draw on published mineral dissolution kinetics, brucite precipitation/dissolution data from prior Stony Brook work by our Ebb collaborators, and PNNL aquarium thresholding experiments characterizing precipitation onset across temperature and salinity. Field-scale calibration draws on Project Macoma observations and on additional alkalinity releases as they come online. Validation focuses on the testbed's ability to reproduce observed alkalinity, DIC, pH, and saturation-state evolution under the corresponding physical and dosing conditions, and its skill in capturing precipitation-onset thresholds and subsequent redissolution patterns.

Task 4. Build ensemble infrastructure for perturbed-parameter runs. Translating the testbed into an uncertainty-quantification engine requires ensemble tooling: orchestration to launch and manage thousands of perturbed-parameter runs, standardized parameter sampling, and analysis pipelines that decompose ensemble output into uncertainty contributions and parameter sensitivities. We will build this infrastructure within C-Star, leveraging existing workflow management tooling. The infrastructure is reusable: any future MARBL-based OAE study, in 1D or 3D, will be able to exploit the same ensemble harness.

Task 5. Conduct perturbed-parameter calculations and attribute uncertainty. We will run perturbed-parameter ensembles across selected combinations of physical regime and dosing strategy. The resulting ensembles characterize uncertainty in net alkalinity production and systematically attribute that uncertainty across process terms. The output is a characterization of the parameter regimes where processes are well-characterized versus those regimes where additional research or measurement effort is needed.

Task 6. Publish scientific results and a documented, tutorial-equipped public release. The testbed and its associated ensemble infrastructure will be released publicly with C-Star, with documentation, tutorials, and worked examples that lower the barrier for community use. Scientific results—calibration against partner and community datasets, regime-resolved uncertainty attribution, and identification of standardization-ready parameter regimes—will be submitted as peer-reviewed publications. We will also produce shorter community-facing technical notes summarizing the practical implications for MRV practitioners.

Why this approach is best-in-class

Three features differentiate this work from existing OAE modeling efforts.

1. **The testbed is built for hypothesis-testing.** A good process model is a comprehensive hypothesis. By stripping physical complexity down to a 1D column, the testbed makes hypothesis testing in the context of empirical constraints tractable, eliminating the confounding influence of 3D turbulent transport. The reduced-complexity model supports rapid iteration, and because the testbed lives natively within ROMS-MARBL, every formulation validated here drops into operational 3D configurations with no porting step.

2. **Uncertainty quantification is the deliverable.** Model calibration/validation is accompanied by a strong focus on uncertainty quantification — answering the questions: How wrong could we be? What processes are most important in particular regimes, and how well known are they? This information provides a basis for research prioritization and can also inform future MRV protocol development.

3. **Infrastructure is the deliverable.** The testbed and its ensemble infrastructure will be released openly within C-Star, with documentation, tutorials, and worked examples that lower the barrier for adoption. We are actively cultivating a C-Star user community — collaborators who can deploy the testbed in new settings, contribute to its development, and advance the corpus of evidence supporting robust quantification of OAE processes.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

Scientific: incomplete process understanding of precipitation dynamics. The controls on mineral precipitation in seawater are poorly characterized and may depend on factors that are difficult to represent in a continuum model, such as background particulate matter and colloidal material acting as nucleation sites. If first-order controls on behavior cannot be parameterized mechanistically, the model may be tunable but will have limited transferability. *Mitigation:* The testbed is explicitly designed to surface this risk. Perturbed-parameter ensembles will quantify the sensitivity of net alkalinity production to precipitation parameters and identify the regimes in which precipitation uncertainty actually matters for crediting outcomes.

Scientific: MARBL inherits assumptions that need relaxing for OAE. Like most ocean BGC models, MARBL tracks total alkalinity and DIC, but not individual cations (Ca^{2+}, Mg^{2+}). Saturation states (Ω_calcite, Ω_aragonite, Ω_brucite) are computed assuming fixed seawater Ca:Mg ratios — an assumption that can break down in the OAE near field. MARBL's carbonate-system equilibrium constants are appropriate for open-ocean conditions but may approach their validity limits in high-pH near-field plumes. *Mitigation:* Explicit cation tracers and an abiotic precipitation pathway are scoped as part of Task 2, with saturation-state calculations refactored to use them. We will evaluate published carbonate chemistry frameworks and implement alternative constant sets as deemed necessary.

Scientific: observations may underdetermine the calibration. The available empirical datasets may underdetermine the parameter space, yielding calibrations with broad posterior uncertainty. *Mitigation:* Uncertainty attribution explicitly characterizes which parameters remain underdetermined and how that uncertainty propagates to net alkalinity outcomes — directly informing where additional theoretical or empirical effort is needed.

Project execution: specialized expertise outside the core team. [C]Worthy has strong modeling and OAE biogeochemistry capacity, but we are likely to desire additional input from experts in particulate dynamics and mineral dissolution. *Mitigation:* The Ebb Carbon subcontract already provides direct access to precipitation expertise, and our budget includes line items for additional external advisors as needed.

Project execution: bandwidth. [C]Worthy is a small team supporting several active projects; personnel bandwidth is the most likely cause of schedule slippage. *Mitigation:* We will hire a postdoc dedicated to scientific work on the project. Tasks are sequenced so that early infrastructure work (Tasks 1–2) is independent of more bandwidth-intensive calibration and ensemble work (Tasks 3–5).

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	1. **A public, open-source 1D OAE process-model testbed within C-Star.** A documented, tutorial-equipped ROMS-MARBL configuration packaged for community use, including the size-resolved sinking-particulate module, abiotic precipitation pathway, cation tracers, and the perturbed-parameter ensemble harness. 2. **Evaluated dissolution and precipitation parameterizations.** Process formulations for mineral-feedstock dissolution and abiotic carbonate precipitation, calibrated against laboratory and field datasets and documented with posterior parameter distributions. 3. **A cation-resolved MARBL extension.** A modification of MARBL's carbonate-chemistry implementation that explicitly tracks Ca^{2+} and Mg^{2+} and computes mineral-specific saturation states consistently with cation availability. Designed for upstream merge into the community MARBL code base. 4. **Uncertainty attribution dataset.** A documented dataset and accompanying analysis describing how

	net-alkalinity-production uncertainty is distributed across dissolution, precipitation, sinking, and aggregation process terms within specific physical regimes and dosing scenarios. 5. **Benchmark configurations and datasets.** Curated datasets from Ebb and other community sources, packaged with the corresponding testbed configurations to enable reproducible benchmarking across the broader OAE modeling community. 6. **Peer-reviewed publications and technical notes.** A methods paper documenting the testbed and its parameterizations; an uncertainty-attribution paper presenting the regime-resolved sensitivity analysis; and shorter technical notes translating findings for MRV practitioners.
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	These deliverables address a central bottleneck blocking OAE from scaling: the field cannot currently quantify near-field alkalinity production with the confidence needed for defensible crediting. **Uncertainty-informed prioritization.** Validated dissolution and precipitation parameterizations,

	combined with uncertainty attribution, inform the expected confidence intervals around net alkalinity production in operational contexts. **Improved MRV protocols.** Identifying parameter regimes where processes are well-characterized — or where their uncertainty contributes negligibly to overall uncertainty budgets — supports standardized treatments across deployments, appropriate crediting uncertainty assessments, and research prioritization. **Accelerated community progress.** The open testbed, ensemble harness, MARBL extensions, and shared benchmark configurations provide a community resource enabling other groups to advance scientific understanding and robust MRV protocols. Elements of the proposed infrastructure are applicable to other open-system CDR methods, including ocean iron fertilization (particulate dynamics) and coastal enhanced weathering.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	**Q1 Foundation.** Stand up the 1D testbed scaffold within C-Star: vertical stratification, parameterized lateral flushing, size-resolved sinking particulates. Initial code release with reproducible spin-up cases. *Milestone:* working 1D scaffold committed to the C-Star repository. **Q2 Chemistry extension.** Implement cation-resolved chemistry in MARBL (Ca, Mg

	passive tracers; stoichiometric coupling to feedstock dissolution and precipitation; saturation-state refactor). *Milestone:* cation-resolved MARBL branch validated against analytical solutions and literature benchmarks. **Q3 Baseline simulations.** Run baseline simulations and establish canonical deployment regimes. Establish the reference configuration for downstream UQ work. *Milestone:* baseline simulations delivered. **Q4 Validation.** Cross-validation against observations from Ebb Carbon's datasets. Iterative refinement of process representations. *Milestone:* validated configuration matched to deployed-system data. **Q5 Validation (cont.).** Submit first manuscript or technical report on testbed architecture and cation-resolved chemistry. *Milestone:* Manuscript or technical report. **Q6 Ensemble UQ.** Perturbed-parameter ensembles to characterize the near-field chemistry within selected deployment regimes. *Milestone:* uncertainty-quantified parameter sensitivities delivered. **Q7 Registry engagement & second manuscript.** Translate results into protocol-relevant findings and uncertainty methods. Submit second manuscript on UQ findings.

	Milestone: synthesis findings delivered. **Q8 Synthesis & public release.** Full open-source release of testbed and parameter products, community webinar, and final reporting. *Milestone:* public release of scientific and protocol-relevant outputs.
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$722,064